M1 Advanced Deep Learning
Project: Ethical Alignment of Small Language Models

1 Objective

The objective of this project is to study and implement ethical alignment methods for small language models. You will compare several alignment strategies under realistic computational constraints.

The project focuses on three complementary approaches:

- Direct Preference Optimization (DPO)
- Reinforcement Learning from Human Feedback (RLHF)
- Retrieval-Augmented Generation (RAG)-based alignment

The goal is not to produce a perfectly aligned model, but to analyze the strengths and limitations of different alignment techniques. The quality of your implementations will be evaluated using the same dataset called ETHICS. You can choose the dataset you want for training, but not use this evaluation dataset for training. Objective is to maximize accuracy on this evaluation benchmark.

2 Evaluation Benchmark

All groups must evaluate on the same benchmark. The official evaluation benchmark for the project is:

ETHICS Dataset

Reference:

Dan Hendrycks et al., "Aligning AI With Shared Human Values", ICLR 2021.

Only use this dataset for testing. You should use a subset of the dataset because it is big, around 100 example by type of ethic tested (commonsense, justice etc...)

3 Project Description

You must start from a publicly available small language model (around 1 to 3B parameters). The project consists of implementing and comparing multiple alignment strategies on a common ethical evaluation benchmark.

Each group must implement at least two of the following approaches.

3.1 Direct Preference Optimization (DPO)

You should fine-tune the model using preference pairs consisting of:

- a preferred answer
- a rejected answer

The objective is to increase the probability of preferred responses while decreasing the probability of rejected responses.

Possible datasets include:

- Anthropic HH-RLHF
- UltraFeedback
- OpenAssistant preference datasets
- Synthetic preference datasets generated by you

3.2 Reinforcement Learning from Human Feedback (RLHF)

You should implement a simplified RLHF pipeline, including

1. supervised fine-tuning (optional)
2. reward model training
3. policy optimization

You may use PPO or lighter alternatives. The emphasis is placed on understanding the pipeline rather than achieving state-of-the-art performance.

3.3 RAG-based Alignment

You should build a retrieval system containing ethical or safety-related guidelines.

At inference time:

1. relevant ethical principles are retrieved
2. the retrieved documents are injected into the prompt
3. the model generates a response conditioned on the retrieved context

Possible retrieval corpora include:

- AI safety guidelines
- constitutions or ethical charters
- policy documents
- manually curated ethical rules

4 Training Data

Each group is free to choose its own alignment data. The training dataset must:

- be clearly documented
- contain a description of preprocessing steps
- specify whether synthetic data was generated
- specify whether external LLMs were used

You should are encouraged to experiment with data quality, prompt design, and preference construction.

5 Evaluation Protocol

You must compare:

- the original baseline model
- the aligned models

The main evaluation metric is accuracy on ethical classification tasks.

6 Implementation Constraints

- Maximum model size: 3B parameters
- Training must fit on consumer GPUs or university servers
- Parameter-efficient fine-tuning methods are encouraged
- LoRA or QLoRA are strongly recommended

7 Expected Deliverables

Each group must submit:

1. a final report in the format of a paper, 8 pages of content plus unlimited references; following ACL format : `https://www.overleaf.com/latex/templates/association-for-computational-l` `jvxskxpnznfj`
2. the complete source code

8 Report Structure

The report should contain:

1. introduction and motivation
2. description of the alignment methods and datasets
3. implementation details and experimental setup
4. qualitative and quantitative results
5. limitations and ethical discussion
6. conclusion

9 References

1. Rafailov et al., "Direct Preference Optimization: Your Language Model is Secretly a Reward Model", NeurIPS 2023.

2. Ouyang et al., "Training Language Models to Follow Instructions with Human Feedback", NeurIPS 2022.

3. Hendrycks et al., "Aligning AI With Shared Human Values", ICLR 2021.

4. Bai et al., "Constitutional AI: Harmlessness from AI Feedback", arXiv 2022.